COMMON ADMISSION TEST & RESERVE BANK OF INDIA APTITUDE

Formula Book: Quantitative Ability & Data Interpretation and Logical Reasoning

High-Yield Formulas, Calculation Shortcuts, and Framework Rules for 96–97th Percentile Execution

SECTION 1: ARITHMETIC FORMULAS (CORE FOUNDATION)

Percentages & Multipliers

• Percentage change = [(Final - Initial) / Initial] * 100
• Successive Percentage Change: Net change = a + b + (ab / 100)%
• Percentage point change vs Percentage change: Moving from 20% to 25% is a 5 percentage point increase, but a 25% increase in relative value.
• Decimal Multiplier: +15% increase = x 1.15; -20% decrease = x 0.80; -37.5% decrease = x (5/8).

Profit, Loss, and Discount

• Cost Price (CP), Selling Price (SP), Marked Price (MP)
• Profit % = [(SP - CP) / CP] * 100; Loss % = [(CP - SP) / CP] * 100
• Discount % = [(MP - SP) / MP] * 100
• Relationship: MP * (1 - Discount%) = CP * (1 + Profit%)
• Faulty Weights: Profit % = [Error / (True Value - Error)] * 100 = [(Goods Claimed - Goods Given) / Goods Given] * 100

Simple and Compound Interest

• Simple Interest: SI = (Principal * Rate * Time) / 100
• Compound Interest: Amount = Principal * (1 + Rate / 100)^Time
• Difference between Compound Interest and Simple Interest for 2 years: Difference = Principal * (Rate / 100)^2
• Difference for 3 years: Difference = Principal * (Rate / 100)^2 * [3 + (Rate / 100)]
• Rule of 72: Doubling time under compounding ≈ 72 / Rate (in percent).

Ratio, Proportion, and Mixtures & Alligation

• If a/b = c/d, then (a+b)/(a-b) = (c+d)/(c-d) [Componendo and Dividendo]
• Alligation Rule: Quantity of Cheaper / Quantity of Dearer = (Price of Dearer - Mean Price) / (Mean Price - Price of Cheaper)
• Removal and Replacement Formula: Final Concentration of Original Liquid = Initial Concentration * [1 - (Replaced Volume / Total Volume)]^n

Time, Speed, and Distance

• Speed = Distance / Time; 1 km/h = 5/18 m/s; 1 m/s = 18/5 km/h
• Average Speed (Constant Distance): 2 * S1 * S2 / (S1 + S2) [Harmonic Mean]
• Average Speed (Constant Time): (S1 + S2) / 2 [Arithmetic Mean]
• Relative Speed: Same direction = |S1 - S2|; Opposite direction = S1 + S2
• Trains: Crossing pole = Distance is Train Length; Crossing platform = Distance is Train Length + Platform Length
• Boats & Streams: Downstream speed = Speed of Boat + Speed of Stream; Upstream speed = Speed of Boat - Speed of Stream
• Circular Tracks: Meeting at starting point = Least Common Multiple of individual lap times; First meeting anywhere = Track Length / Relative Speed

Time and Work, Pipes and Cisterns

• If A finishes work in X days and B in Y days: Combined time = (X * Y) / (X + Y)
• Man-Days Formula: (Men1 * Days1 * Hours1) / Work1 = (Men2 * Days2 * Hours2) / Work2
• Efficiency: Efficiency is inversely proportional to time taken when work is constant.
• Negative Work: Inflow pipe rate (+1/X), Outflow emptying pipe rate (-1/Y).

SECTION 2: ALGEBRA FORMULAS (HIGH-WEIGHTAGE)

Quadratic and Higher-Degree Equations

• Standard Quadratic: a*x^2 + b*x + c = 0; Roots: x = [-b ± sqrt(b^2 - 4*a*c)] / (2*a)
• Sum of roots = -b/a; Product of roots = c/a
• Discriminant (D = b^2 - 4*a*c): D > 0 (real and distinct), D = 0 (real and equal), D < 0 (complex conjugate roots)
• If roots are rational and coefficients are rational, D must be a perfect square.
• Maximum/Minimum of Quadratic: For a > 0, minimum occurs at x = -b / (2*a), value = (4*a*c - b^2) / (4*a). For a < 0, maximum occurs at x = -b / (2*a).

Inequalities and Modulus

• Modulus Property: |x| ≤ a means -a ≤ x ≤ a; |x| ≥ a means x ≥ a or x ≤ -a
• Triangle Inequality: |a + b| ≤ |a| + |b|
• Arithmetic Mean ≥ Geometric Mean ≥ Harmonic Mean (equality holds when all terms are equal)
• Wavy Curve Method (Sign Scheme): Factorize expression, plot critical zeros on number line, determine sign alternation.

Functions and Logarithms

• Logarithm Definition: log_b(a) = c implies b^c = a (where a > 0, b > 0, b ≠ 1)
• Product Rule: log(x * y) = log(x) + log(y); Quotient Rule: log(x / y) = log(x) - log(y)
• Power Rule: log(x^k) = k * log(x); Base Change Rule: log_b(a) = log_c(a) / log_c(b)
• Reciprocal Base: log_b(a) = 1 / log_a(b); Power Base: log_(b^k)(a) = (1/k) * log_b(a)

Progressions and Series

• Arithmetic Progression: n-th term = a + (n - 1)*d; Sum = (n / 2) * [2*a + (n - 1)*d]
• Geometric Progression: n-th term = a * r^(n - 1); Sum = a * (r^n - 1) / (r - 1)
• Infinite Geometric Progression (|r| < 1): Sum = a / (1 - r)
• Standard Sums: Sum of first n naturals = n*(n + 1) / 2; Sum of squares = n*(n + 1)*(2*n + 1) / 6; Sum of cubes = [n*(n + 1) / 2]^2

SECTION 3: MODERN MATHEMATICS & NUMBER SYSTEMS

Permutations, Combinations & Probability

• Permutations (Arrangements): nPr = n! / (n - r)!
• Combinations (Selections): nCr = n! / [r! * (n - r)!]; Note: nCr = nC(n-r)
• Circular Permutations: (n - 1)! for distinct objects; (n - 1)! / 2 for necklaces/keyrings
• Identical Objects Distribution: Number of ways to distribute n identical items among r distinct people:
- Any number of items (including zero): (n + r - 1) C (r - 1)
- At least one item each: (n - 1) C (r - 1)
• Probability = Favorable Outcomes / Total Outcomes; P(A or B) = P(A) + P(B) - P(A and B)
• Independent Events: P(A and B) = P(A) * P(B)

Number Systems & Remainder Theorems

• Prime Factorization: N = p^a * q^b * r^c
• Number of Factors: Total factors = (a + 1) * (b + 1) * (c + 1)
• Sum of Factors = [(p^(a+1) - 1) / (p - 1)] * [(q^(b+1) - 1) / (q - 1)] * [(r^(c+1) - 1) / (r - 1)]
• Number of Coprimes less than N (Euler's Totient): phi(N) = N * (1 - 1/p) * (1 - 1/q) * (1 - 1/r)
• Euler's Theorem: If a and m are coprime, then [a^phi(m)] divided by m leaves remainder 1.
• Fermat's Little Theorem: If p is prime, [a^(p - 1)] divided by p leaves remainder 1.

SECTION 4: GEOMETRY & MENSURATION

Triangles, Circles, and Polygons

• Area of Triangle = (1/2) * Base * Height = sqrt[s * (s - a) * (s - b) * (s - c)] where s = (a + b + c)/2
• Inradius (r) = Area / s; Circumradius (R) = (a * b * c) / (4 * Area)
• Similar Triangles: Ratio of areas = (Ratio of corresponding sides)^2
• Circles: Area = pi * r^2; Circumference = 2 * pi * r; Tangent-Secant Theorem: PT^2 = PA * PB
• Polygons: Sum of interior angles = (n - 2) * 180°; Number of diagonals = n * (n - 3) / 2

3D Solids Mensuration

• Cylinder: Volume = pi * r^2 * h; Curved Surface Area = 2 * pi * r * h; Total Surface Area = 2 * pi * r * (r + h)
• Cone: Volume = (1/3) * pi * r^2 * h; Slant height l = sqrt(r^2 + h^2); Curved Surface Area = pi * r * l
• Sphere: Volume = (4/3) * pi * r^3; Surface Area = 4 * pi * r^2
• Hemisphere: Volume = (2/3) * pi * r^3; Total Surface Area = 3 * pi * r^2

SECTION 5: DATA INTERPRETATION & LOGICAL REASONING RULES

Data Interpretation Calculation Rules

• Compounded Annual Growth Rate Approximation: CAGR ≈ [(Final Value / Initial Value)^(1/n) - 1]
• Faster Approximation: CAGR ≈ (Overall Percentage Growth / Number of Years) * Correction Factor
• Percentage Comparison Shortcuts: To compare a/b vs c/d, check cross-multiplication a*d vs b*c.
• Weighted Average Formula: Combined Average = (N1*X1 + N2*X2) / (N1 + N2)

Logical Reasoning Execution Frameworks

• Seating Arrangements: Always identify the 'Anchor Clue' (clue connecting person to fixed position like corner or extreme left/right). Never branch more than 2 sub-cases.
• Multi-Attribute Grids: Create a grid with the most constrained entity (e.g., people vs days vs professions). Mark definite cross (X) and ticks.
• Knockout Tournaments: In a tournament of N players with standard seeding (Seed 1 plays Seed N, Seed 2 plays Seed N-1), sum of match seeds in Round 1 = N + 1. Total matches played to determine winner = N - 1.
• Routes and Networks: Work backwards from the destination node to the origin node. Calculate cumulative inward and outward capacities.